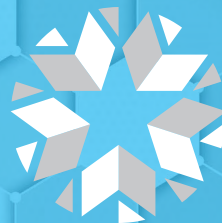


# **OKLAHOMA SCHOOL TESTING PROGRAM**

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TEST BLUEPRINT AND ITEM SPECIFICATIONS  
**GRADE 6 MATHEMATICS**



**OKLAHOMA**  
Education

**1**

Drag the integers into the spaces in order from least to greatest. To drag an integer, click and hold the integer, and then drag it to the desired space. To change an integer, click and hold it, and then drag it back to the desired space.

|             |   |             |   |             |   |             |
|-------------|---|-------------|---|-------------|---|-------------|
| <div></div> | < | <div></div> | < | <div></div> | < | <div></div> |
|-------------|---|-------------|---|-------------|---|-------------|

7

-8

0

-6

**Standard: 6.N.1.2** Compare and order positive rational numbers, represented in various forms, or integers using the symbols  $<$ ,  $>$ , and  $=$ .

**Depth-of-Knowledge: 1**

This item is a DOK 1 because it requires the student to complete a simple procedure, ordering numbers from smallest to largest.

**Sample Distractor Rationales:**

**Correct**

$-8 < -6 < 0 < 7$

**Incorrect**

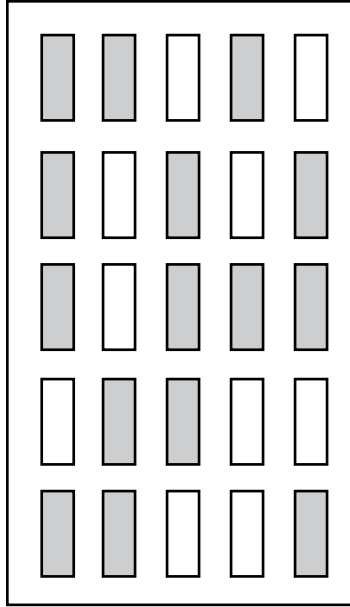
$0 < -6 < 7 < -8$

The student ignored the negative signs.

$-6 < -8 < 0 < 7$

The student knew that the negative numbers were smaller, but didn't know how to order  $-6$  and  $-8$ .

- 2** The office building shown has shaded windows for the rooms in which the lights are turned off.



**What portion of the rooms have their lights turned off?**

- A**  $\frac{2}{3}$
- B**  $\frac{3}{4}$
- C** 15%
- D** 60%

**Standard: 6.N.1.4** Determine equivalencies among fractions, mixed numbers, decimals, and percents.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to find the number of rooms with lights turned off and then express this as a fraction.

**Distractor Rationale:**

- A. The student saw that on the top or bottom floor there are 2 unshaded windows and 3 shaded windows.
- B. Balance distractor
- C. The student found the number of shaded windows, 15, and thought this meant 15%.
- D. **Correct.** The student demonstrated an ability to determine equivalency among a fraction and a percent to solve a problem.

- 3** Eli's family had \$60 for a party. They spent \$28 on food. They were given an additional \$10. This expression shows how much money, in dollars, Eli's family has available for the party.

$$60 + ( - 28 ) + 10$$

**How much money does Eli's family have available for the party?**

- A** \$22
- B** \$42
- C** \$58
- D** \$98

**Standard: 6.N.2.3** Add and subtract integers in a variety of situations; use efficient and generalizable procedures including but not limited to standard algorithms.

**Depth-of-Knowledge: 1**

This item is a DOK 1 because it requires the student to complete a simple procedure, adding and subtracting integers.

**Distractor Rationale:**

- A. The student subtracted 10 instead of added 10.
- B. Correct. The student demonstrated an ability to add and subtract integers.**
- C. Balance distractor
- D. The student ignored the negative sign on 28.

- 4** The temperature at a location was  $-12^{\circ}$  Fahrenheit ( $^{\circ}\text{F}$ ). The table shows the changes in temperature over the next 4 hours.

**Changes in Temperature**

| Hour | Change in Temperature ( $^{\circ}\text{F}$ ) |
|------|----------------------------------------------|
| 1    | 4                                            |
| 2    | $-3$                                         |
| 3    | $-5$                                         |
| 4    | 3                                            |

**What is the temperature at the end of the 4th hour?**

- A**  $-13^{\circ}\text{F}$
- B**  $-11^{\circ}\text{F}$
- C**  $11^{\circ}\text{F}$
- D**  $13^{\circ}\text{F}$

**Standard: 6.N.2.3** Add and subtract integers in a variety of situations; use efficient and generalizable procedures including but not limited to standard algorithms.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to extract information from the table and then add and subtract integers.

**Distractor Rationale:**

- A. Correct.** The student demonstrated an ability to add and subtract integers.
- B.** Balance distractor
- C.** The student missed the negative on  $-12$ .
- D.** The student computed  $-12 + 4 - 3 - 5 + 3 = -13$ , but then forgot the negative sign for the answer.

A student has a box of crayons. The table shows the color and number of crayons in the box.

**Crayons in a Box**

| Color  | Number |
|--------|--------|
| Blue   | 5      |
| Red    | 9      |
| Green  | 7      |
| Yellow | 1      |

**Select the color or ratio that best completes each statement.** To select a word or color, click the menu and then click the desired color or ratio. To choose a different color or ratio, click the menu and click the new color or ratio.

Based on the information in the table, the ratio of  
 -Select an Answer- crayons to -Select an Answer-  
 crayons in the box is -Select an Answer-.

Based on the information in the table, the ratio of

blue crayons to -Select an Answer-  
 blue s -Select an Answer-  
 red  
 green  
 yellow

Based on the information in the table, the ratio of

-Select an Answer- crayons to -Select an Answer-  
 crayons in the box is -Select an Answer-  
 blue  
 red  
 green  
 yellow

Based on the information in the table, the ratio of

-Select an Answer- crayons to -Select an Answer-  
 crayons in the box is -Select an Answer-  
 1 to 9  
 5 to 8  
 7 to 6  
 14 to 1

**Standard: 6.N.3.1** Identify and use ratios to compare and relate quantities in multiple ways. Recognize that multiplicative comparison and additive comparison are different.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to compare many different colors of crayons using ratio.

### Sample Distractor Rationales:

#### Correct

Based on the information in the table, the ratio of  
yellow crayons to red  
crayons in the box is 1 to 9.

#### Incorrect

Based on the information in the table, the ratio of blue crayons to red crayons in the box is: 5 to 8.

The student found the correct number for blue, but chose an incorrect number for red.

Based on the information in the table, the ratio of green crayons to yellow crayons is: 1 to 9.

The student did not understand that the order of the crayons in the ratio is important.

Based on the information in the table, the ratio of green crayons to yellow crayons is: 7 to 6.

The student's ratio showed green crayons to green minus yellow crayons.

- 6** In a survey of 292 students, about 9.9% have attended more than one play.
- Which is **closest** to the number of students in the survey who have attended more than one play?
- A** 3 students
  - B** 10 students
  - C** 20 students
  - D** 30 students

**Standard: 6.N.3.3** Apply the relationship between ratios, equivalent fractions, unit rates, and percents to solve problems in various contexts.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to determine how to solve the problem and then find the closest, not the exact, answer.

**Distractor Rationale:**

- A. The student confused 10% and 1%
- B. The student thought 10% was the same as 10 students.
- C. The student rounded 292 to 200.
- D. **Correct.** The student demonstrated an ability to apply the relationship between ratios and percents to solve a problems with a real-world context.

**7** During soccer practice, the goalie blocked 72% of the shots attempted by the opponents. If the goalie blocked a total of 18 shots, how many total shots were attempted?

- A** 5
- B** 13
- C** 20
- D** 25

**Standard: 6.N.3.3** Apply the relationship between ratios, equivalent fractions, unit rates, and percents to solve problems in various contexts.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to decide how to approach the problem, finding the total number of shots attempted when given the percentage and number of shots blocked.

**Distractor Rationale:**

- A. The student found 28% of 18, then rounded to the nearest whole number.
- B. The student found 72% of 18, then rounded to the nearest whole number.
- C. Balance distractor
- D. Correct. The student demonstrated an ability to apply the relationships between ratios, equivalent fractions, and percents to solve a real-world problem.

- 8** A chef ordered \$138 worth of groceries. The chef had a coupon for \$14 off the order.

Which estimate is **closest** to the fraction of the cost saved using this coupon?

**A**  $\frac{1}{100}$

**B**  $\frac{10}{100}$

**C**  $\frac{14}{100}$

**D**  $\frac{20}{100}$

**Standard: 6.N.4.1** Estimate solutions to problems with whole numbers, decimals, fractions, and mixed numbers and use the estimates to assess the reasonableness of results in the context of the problem.

**Depth-of-Knowledge: 2**

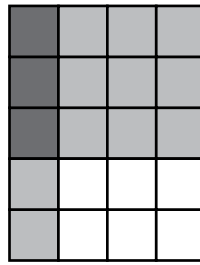
This item is a DOK 2 because it requires the student to interpret the scenario and then find the percent of money saved.

**Distractor Rationale:**

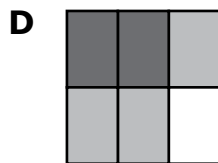
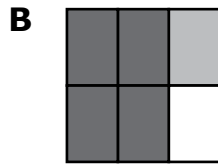
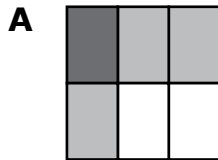
- A. The student made a place value error.
- B. Correct. The student demonstrated an ability to estimate the solution to a problem using a fraction.**
- C. The student used the discount as the numerator.
- D. Balance distractor

9

This model represents the product of  $\frac{1}{4} \cdot \frac{3}{5}$



Which model correctly represents the product of  $\frac{2}{3} \cdot \frac{1}{2}$ ?



**Standard: 6.N.4.2** Illustrate multiplication and division of fractions and decimals to show connections to fractions, whole number multiplication, and inverse relationships.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to translate a multiplication expression into a model.

**Distractor Rationale:**

A. Balance distractor

B. Student saw  $\frac{1}{2}$  of the whole shaded on the top and  $\frac{2}{3}$  of the bottom shaded.

C. Student focused on denominators and saw 3 shaded on the bottom and 2 shaded on the top.

D. **Correct.** The student demonstrated an ability to illustrate multiplication of fractions using a model.

**10**

**A  $4\frac{1}{2}$ -pound bag of jellybeans is shared equally by 6 friends. What amount of jellybeans does each friend get?**

- A**  $\frac{1}{2}$  pound
- B**  $\frac{3}{4}$  pound
- C**  $1\frac{1}{6}$  pounds
- D**  $1\frac{1}{3}$  pounds

**Standard: 6.N.4.3** Multiply and divide fractions and decimals using efficient and generalizable procedures.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because the student must use a strategy to divide a mixed number by a whole number, resulting in a fractional answer.

**Distractor Rationale:**

- A. The student chose a denominator of 2 because the original mixed number had a denominator of 2.
- B. Correct. The student demonstrated an ability to divide a mixed number by a whole number.**
- C. The student chose a denominator of 6 because the jellybeans will be shared equally by 6 friends.
- D. Balance distractor

- 11** Mr. Lopez bought several types of meat for a party. The amount, in pounds, of each type he bought is shown in the table.

**Meats**

| Type       | Amount (pounds) |
|------------|-----------------|
| ham        | 2.53            |
| pastrami   | 0.44            |
| turkey     | 3.61            |
| roast beef | 1.49            |
| salami     | 1.92            |

Which is closest to the total amount of meat Mr. Lopez bought?

- A 7 pounds
- B 8 pounds
- C 10 pounds
- D 12 pounds

**Standard: 6.N.4.4** Use mathematical modeling to solve and interpret problems including money, measurement, geometry, and data requiring arithmetic with decimals, fractions and mixed numbers.

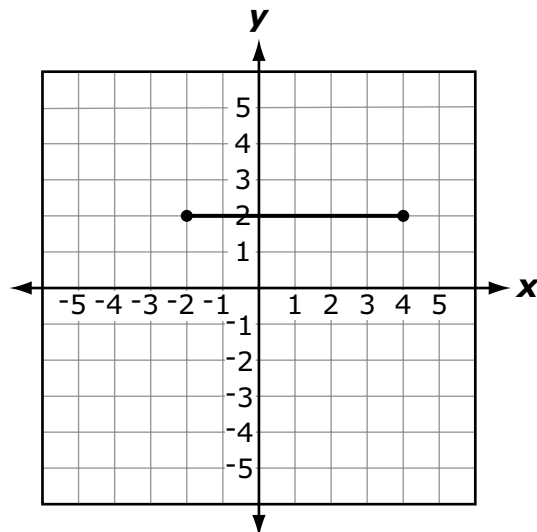
**Depth-of-Knowledge: 2**

This item is a DOK 2 because the student must extract the information from the table and then find the closest, not the exact, answer.

**Distractor Rationale:**

- A. The student ignored the decimals and added the numbers in the ones places only.
- B. The student ignored the decimals and added the numbers in the ones places only and then added 1 more to account for the decimals.
- C. **Correct.** The student demonstrated an ability to estimate the solution to an addition of integers problem.
- D. The student rounded each amount up to the next whole number and then added.

- 12** One side of a rectangle is shown on this graph.



Which coordinates could be the missing vertices of the rectangle?

- A**  $(-5, -2)$  and  $(-5, 4)$
- B**  $(2, -2)$  and  $(2, 4)$
- C**  $(-2, 3)$  and  $(4, -3)$
- D**  $(-2, -5)$  and  $(4, -5)$

**Standard: 6.A.1.1** Plot integer- and rational-valued (limited to halves and fourths) ordered-pairs as coordinates in all four quadrants and recognize the reflective relationships among coordinates that differ only by their signs.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to think about where the other vertices of a rectangle could be and then choose the coordinates of the possible vertices.

**Distractor Rationale:**

- A. The student reversed the x- and y-coordinates.
- B. The student gave the coordinates of the given side of the rectangle and reversed the x- and y-coordinates.
- C. The student focused on the x-coordinates only.
- D. Correct. The student demonstrated an ability to plot integer ordered pairs in all four quadrants.**

- 13** The table shows the total number of pictures Cal took by the end of each week.

**Cal's Pictures**

| <b>Week<br/>(<math>w</math>)</b> | <b>Total<br/>Number of<br/>Pictures</b> |
|----------------------------------|-----------------------------------------|
| 1                                | 4                                       |
| 2                                | 8                                       |
| 3                                | 12                                      |
| 4                                | 16                                      |

**Based on this pattern, which expression can be used to find the total number of pictures Cal took by the end of  $w$  weeks?**

- A**  $2 \cdot w$
- B**  $4 \cdot w$
- C**  $w + 12$
- D**  $4 \cdot w + 4$

**Standard: 6.A.1.2** Represent relationships between two varying positive quantities involving no more than two operations with rules, graphs, and tables; translate between any two of these representations.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to first figure out the pattern presented in a table and then explain the relationship between terms in the pattern using an algebraic expression.

**Distractor Rationale:**

- A. The student saw that the total number of pictures doubled from week 1 to week 2.
- B. Correct. The student demonstrated an ability to represent a real-world situation using an expression involving a variable.**
- C. The student added the values from week 1 and week 2 to get 12.
- D. The student thought you had to add the 4 from week 1.

**14** Ms. Jones wrote this rule on the board.

**"a number,  $n$ , increased by eighteen"**

**Which expression represents the rule Ms. Jones wrote on the board?**

**A**  $n + 18$

**B**  $n - 18$

**C**  $n \cdot 18$

**D**  $n \div 18$

**Standard: 6.A.1.3** Use and evaluate variables in expressions, equations, and inequalities that arise from various contexts, including determining when or if, for a given value of the variable, an equation or inequality involving a variable is true or false.

**Depth-of-Knowledge: 1**

This item is a DOK 1 because it requires the student to complete a simple procedure, turning a expression given in words into an algebraic expression.

**Distractor Rationale:**

- A. **Correct.** The student demonstrated an ability to write an expression with a variable from a rule given in words.
- B. The student used the wrong operation.
- C. The student used the wrong operation.
- D. The student used the wrong operation.

**15** What is the value of this expression?

$$10 \div (2 + 3) \bullet 5 - 3$$

- A** 4
- B** 7
- C** 17
- D** 37

**Standard: 6.A.2.1** Generate equivalent expressions and evaluate expressions involving positive rational numbers by applying the commutative, associative, and distributive properties and order of operations to model and solve mathematical problems.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to evaluate a multi-step expression using the order of the operations.

**Distractor Rationale:**

- A. Balance distractor
- B. Correct.** The student demonstrated an ability to evaluate an expression using positive rational numbers by applying the order of operations.
- C. The student computed  $(10 \div 2) + (3 \times 5) - 3$ .
- D. The student computed from left to right, ignoring the order of operations.

**16** What is the value of this numerical expression?

$$3.27 + 4.06 \bullet 2 - (3.19 - 0.18)$$

- A** 11.65
- B** 11.29
- C** 8.38
- D** 8.02

**Standard: 6.A.2.1** Generate equivalent expressions and evaluate expressions involving positive rational numbers by applying the commutative, associative, and distributive properties and order of operations to model and solve mathematical problems.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to evaluate a multi-step expression using the order of the operations.

**Distractor Rationale:**

- A. The student computed what was in the parenthesis first and then computed from left to right ignoring the order of operations.
- B. The student computed from left to right, ignoring the order of operations and the parenthesis.
- C. Correct. The student demonstrated an ability to evaluate an expression using the order of operations.**
- D. The student performed the multiplication first and then computed from left to right, ignoring the order of operations.

**Match the expression in the left column to each equivalent expression in the right column.** To connect expressions, click an expression in the left column and then an expression in the right column, and a line will automatically be drawn between them. To remove a connection, hold the pointer over the line until it turns red, and then click it. Each expression in the left column matches to only one expression in the right column.

|                   |                 |
|-------------------|-----------------|
| $5(6 + 8)$        | $40 + 48$       |
| $48 + 30$         | $8 + 30$        |
| $(5 + 6) \cdot 8$ | $6(5 + 8)$      |
| $5 + 48$          | $40 + 30$       |
| $6 \cdot 5 + 8$   | $5 + 6 \cdot 8$ |

**Standard: 6.A.2.1** Generate equivalent expressions and evaluate expressions involving positive rational numbers by applying the commutative, associative, and distributive properties and order of operations to model and solve mathematical problems.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to find equivalent representations of expressions using different properties.

**Sample Distractor Rationales:**

**Correct**

Diagram showing correct matching between expressions in the left and right columns:

- $5(6 + 8)$  is connected to  $40 + 48$ .
- $48 + 30$  is connected to  $8 + 30$ .
- $(5 + 6) \cdot 8$  is connected to  $6(5 + 8)$ .
- $5 + 48$  is connected to  $40 + 30$ .
- $6 \cdot 5 + 8$  is connected to  $5 + 6 \cdot 8$ .

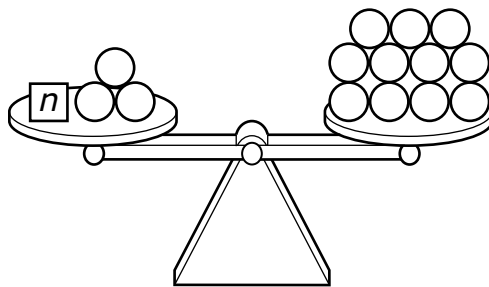
**Incorrect**

Diagram showing incorrect matching between expressions in the left and right columns:

- $5(6 + 8)$  is connected to  $8 + 30$ .
- $48 + 30$  is connected to  $40 + 48$ .
- $(5 + 6) \cdot 8$  is connected to  $6(5 + 8)$ .
- $5 + 48$  is connected to  $40 + 30$ .
- $6 \cdot 5 + 8$  is connected to  $5 + 6 \cdot 8$ .

The student did not know how to apply the distributive property.

**18** A student balanced a scale as shown.



How many s equal  $n$ ?

- A 3
- B 8
- C 11
- D 14

**Standard: 6.A.3.1** Model mathematical situations using expressions, equations, and inequalities involving variables and rational numbers.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to use a balance scale to determine the value of one object in relation to another object.

**Distractor Rationale:**

- A. The student focused on the 3 circles on the left side of the balance scale.
- B. **Correct.** The student demonstrated an ability to represent a mathematical situation using an equation.
- C. The student added the 8 circles on the right to the 3 on the left.
- D. The student added the 8 circles on the right to the 3 on the left and then added 3 more for the value of  $n$ .

**19** A student survey showed there were more dog owners than cat owners. There were 21 cat owners in the survey. If  $d$  represents the number of dog owners, which inequality shows the relationship between the number of dog and cat owners in this survey?

- A**  $d > 21$
- B**  $d \geq 21$
- C**  $d < 21$
- D**  $d \leq 21$

**Standard: 6.A.3.1** Model mathematical situations using expressions, equations, and inequalities involving variables and rational numbers.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to translate a verbal description of a real-life situation into an inequality.

**Distractor Rationale:**

- A. Correct.** The student demonstrated an ability to represent a real-world situation using an expression involving a variable.
- B.** The student thought that the number of dog owners could be the same as the number of cat owners or did not understand what the greater than or equal to symbol means.
- C.** The student confused the less than and greater than symbols.
- D.** The student confused the less than and greater than symbols and thought that the number of dog owners could be the same as the number of cat owners or did not understand what the less than or equal to symbol means.

- 20** Carly has \$10. She used this equation to determine how many tickets,  $n$ , she can buy.

$$2 \bullet n = 10$$

**How many tickets can Carly buy?**

- A** 5 tickets
- B** 8 tickets
- C** 12 tickets
- D** 20 tickets

**Standard: 6.A.3.2** Use number sense and properties of operations and equality to model and solve mathematical problems involving equations in the form  $x + p = q$  and  $px = q$ , where  $p$  and  $q$  are nonnegative rational numbers. Graph the solution on a number line, interpret the solution in the original context, and assess the reasonableness of the solution.

**Depth-of-Knowledge: 1**

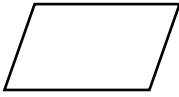
This item is a DOK 1 because it requires the student to complete a simple procedure, solving for a variable in an equation.

**Distractor Rationale:**

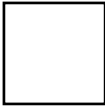
- A. Correct.** The student demonstrated an ability to solve a real-world problem involving an equation with one variable.
- B.** The student computed  $10 - 2$ .
- C.** The student computed  $10 + 2$ .
- D.** The student computed  $10 \times 2$ .

**21** Which shape has no lines of symmetry?

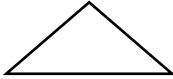
**A**



**B**



**C**



**D**



**Standard: 6.GM.1.3** Identify and describe the line(s) of symmetry in two-dimensional shapes.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to assess all shapes and find the one with no lines of symmetry.

**Distractor Rationale:**

- A. Correct. The student demonstrated an ability to identify lines of symmetry in two-dimensional shapes.
- B. Balance distractor
- C. The student saw that there is no horizontal line of symmetry.
- D. The student thought that only polygons can have lines of symmetry.

**22** Mrs. Thompson has some square tiles. If each side measures 5 inches, what is the area of 1 square tile?

- A** 25 square inches
- B** 20 square inches
- C** 10 square inches
- D** 5 square inches

**Standard: 6.GM.2.1** Develop and use formulas for the area of squares and parallelograms using a variety of methods including but not limited to the standard algorithms and finding unknown measures.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to know that squares have all sides that measure the same length and then find the area of a square with a given side length.

**Distractor Rationale:**

- A. Correct.** The student demonstrated an ability to use a formula to find the area of a square.
- B.** The student found the perimeter.
- C.** The student computed  $5 + 5$ .
- D.** The student computed  $5 \times 1$ .

**23** Two architects designed an object in the shape of an equilateral triangle. The height of the object is 13.5 meters, and the side length is 15.6 meters. What is the area, in square meters, of the triangular object?

- A** 315.9
- B** 210.6
- C** 105.3
- D** 46.8

**Standard: 6.GM.2.2** Develop and use formulas to determine the area of triangles and find unknown measures.

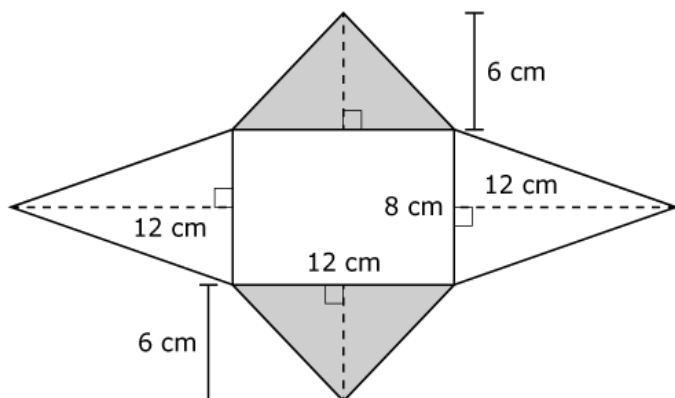
**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to understand that the length of an equilateral triangle is the measure of the base and use this to find the area of a triangle.

**Distractor Rationale:**

- A. The student computed  $1\frac{1}{2} \times \text{base} \times \text{height}$ .
- B. The student computed  $\text{base} \times \text{height}$ .
- C. Correct.** The student demonstrated an ability to find the area of a triangle.
- D. The student computed  $3 \times \text{base}$ .

Antoine created the figure shown using four isosceles triangles and one rectangle.



**Select the number for each measure to complete the sentences.** To select a number, click the menu and then click the desired number. To choose a different number, click the menu and click the new number.

The area of each shaded triangle is  square centimeters ( $\text{cm}^2$ ).

- 9
- 18
- 36
- 72

The area of each unshaded triangle is  square centimeters ( $\text{cm}^2$ ).

- 20
- 40
- 48
- 96

The total area of the figure is  square centimeters ( $\text{cm}^2$ ).

- 40
- 48
- 96
- 180
- 264

**Standard: 6.GM.2.3** Find the area of right triangles, other triangles, special quadrilaterals, and polygons that can be decomposed into triangles and other shapes.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to find the area of a composite figure.

**Sample Distractor Rationales:****Correct**

The area of each shaded triangle is 36 square centimeters ( $\text{cm}^2$ ).

The area of each unshaded triangle is 48 square centimeters ( $\text{cm}^2$ ).

The total area of the figure is 264 square centimeters ( $\text{cm}^2$ ).

**Incorrect**

The area of each shaded triangle is 72 square centimeters ( $\text{cm}^2$ ).

The student used  $A = B \times h$  for the area of a triangle.

The area of each unshaded triangle is 96 square centimeters ( $\text{cm}^2$ ).

The student used  $A = B \times h$  for the area of a triangle.

The total area of the figure is 264 square centimeters ( $\text{cm}^2$ ).

The student used  $A = B \times h$  for the area of a triangle and then only found the area of the rectangle and two triangles.

The area of each shaded triangle is 18 square centimeters ( $\text{cm}^2$ ).

The student found the area of half of the triangle because the triangle is divided by a dashed line.

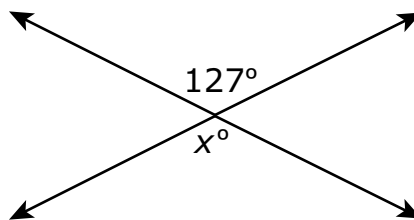
The area of each unshaded triangle is 24 square centimeters ( $\text{cm}^2$ ).

The student found the area of half of the triangle because the triangle is divided by a dashed line.

The total area of the figure is 80 square centimeters ( $\text{cm}^2$ ).

The student found the area of half of the triangles because the triangles are divided by dashed lines.

**25** Two lines intersect in the diagram shown below.



**What is the value of  $x$ ?**

- A** 37
- B** 53
- C** 127
- D** 217

**Standard: 6.GM.3.1** Solve problems using the relationships between the angles (vertical, complementary, and supplementary) formed by intersecting lines.

**Depth-of-Knowledge: 1**

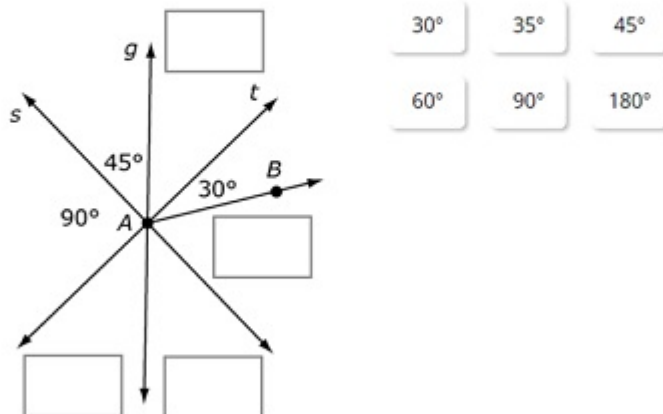
This item is a DOK 1 because it requires the student to recall the definition of vertical angles.

**Distractor Rationale:**

- A. The student computed  $180 - 127 = 53$  and then  $90 - 53 = 37$ .
- B. The student thought the two angles were supplementary.
- C. **Correct.** The student demonstrated an ability to use the relationships between angles formed by intersecting lines to identify an angle measure.
- D. The student thought the difference of the two angles must be 90.

Lines  $s$ ,  $g$ ,  $t$ , and ray  $AB$  intersect at point  $A$ .

**Drag and drop angle measures to show the measures of all the angles.** To place an angle measure in the figure, click and hold the angle measure, and then drag it to the desired space. To change an angle measure, click and hold it, and then drag it back to the desired space. You may use each angle once, more than once, or not at all.



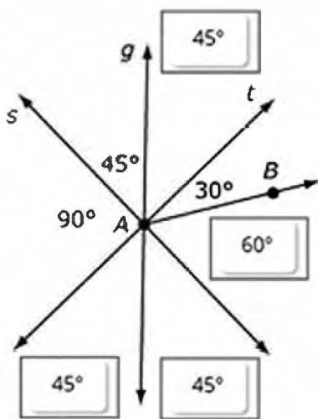
**Standard: 6.GM.3.1** Solve problems using the relationships between the angles (vertical, complementary, and supplementary) formed by intersecting lines.

**Depth-of-Knowledge: 2**

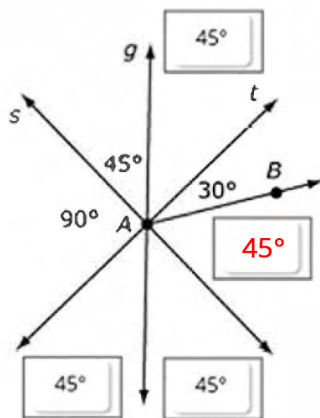
This item is a DOK 2 because it requires the student to use knowledge about the relationships between the angles formed by intersecting lines and use this to identify the angle measures.

**Sample Distractor Rationales:**

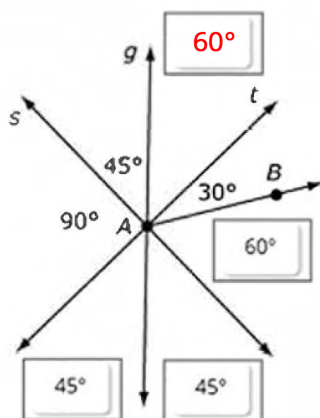
Correct



Incorrect

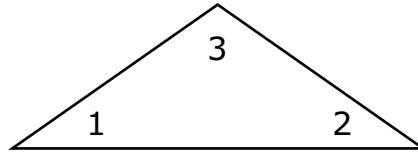


The student thought all missing angle measurements appeared congruent.



The student thought the incorrect angle was complementary to the 30-degree angle.

**27** In the triangle shown,  $m\angle 1 = m\angle 2$  and  $m\angle 2 = 35^\circ$ .



**Which statement must be true?**

- A**  $m\angle 1 + m\angle 3 = 35^\circ$
- B**  $m\angle 1 + m\angle 3 = 70^\circ$
- C**  $m\angle 1 + m\angle 3 = 145^\circ$
- D**  $m\angle 1 + m\angle 3 = 180^\circ$

**Standard: 6.GM.3.2** Develop and use the fact that the sum of the interior angles of a triangle is  $180^\circ$  to determine missing angle measures in a triangle.

**Depth-of-Knowledge: 2**

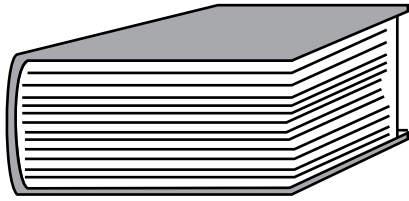
This item is a DOK 2 because it requires the student to use the knowledge that the sum of the interior angles of a triangle is  $180^\circ$  and use this to solve for unknown angle measures in a triangle.

**Distractor Rationale:**

- A. The student chose because 35 degrees is part of the givens.
- B. The student confused angle 2 and angle 3.
- C. Correct. The student demonstrated an ability to use the fact that the sum of the interior angles of a triangle is  $180^\circ$  to determine the missing angle measures in a triangle.**
- D. The student thought that the sum of two angles of a triangle equals  $180^\circ$ .

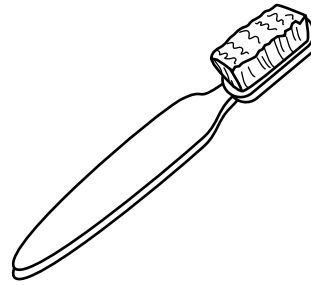
**28** Which object weighs about 1 ounce?

**A**



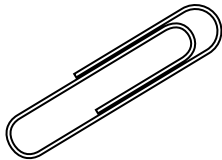
**Book**

**B**



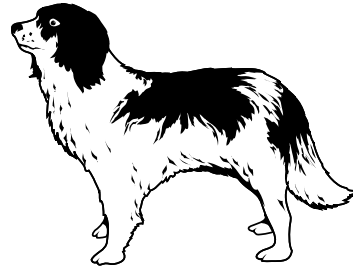
**Toothbrush**

**C**



**Paper clip**

**D**



**Dog**

**Standard: 6.GM.4.1** Estimate weights and capacities using benchmarks in customary and metric measurement systems with appropriate units.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to estimate weights.

**Distractor Rationale:**

- A. The student did not understand how much 1 ounce is.
- B. Correct. The student demonstrated an ability to estimate weight in ounces.**
- C. The student chose an object that weighs the least.
- D. The student did not understand how much 1 ounce is.

**29** A centimeter is equal to 10 millimeters. If a cell phone is 5 centimeters wide, what is the width of the cell phone in millimeters?

- A** 2
- B** 10
- C** 20
- D** 50

**Standard: 6.GM.4.2** Solve problems that require the conversion of lengths within the same measurement systems using appropriate units.

**Depth-of-Knowledge: 2**

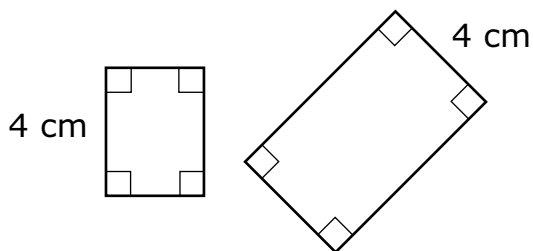
This item is a DOK 2 because it requires the student to convert between centimeters and millimeters.

**Distractor Rationale:**

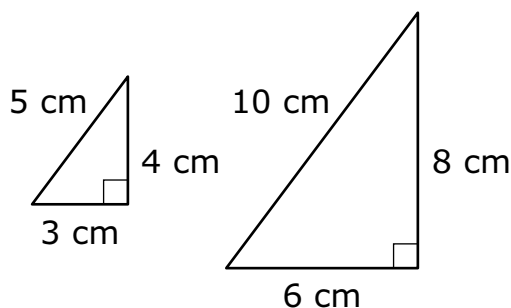
- A. The student computed  $10 \div 5$ .
- B. The student thought millimeters and centimeters are the same size or chose the conversion factor.
- C. The student computed  $10 \times 2$ .
- D. Correct. The student demonstrated an ability to solve a real-world problem that requires the conversion from centimeters to millimeters.**

**30** Which of these **best** represents a pair of congruent shapes?

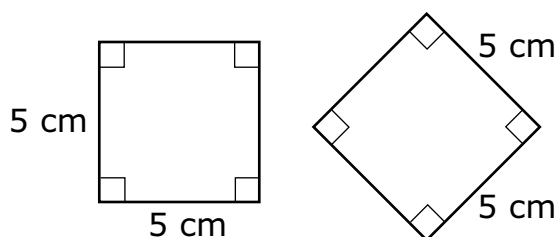
**A**



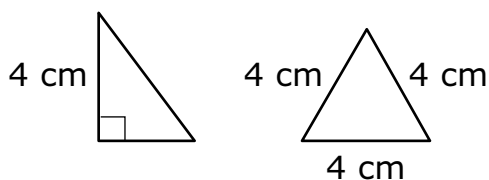
**B**



**C**



**D**



**Standard: 6.GM.4.2** Solve problems that require the conversion of lengths within the same measurement systems using appropriate units.

**Depth-of-Knowledge: 1**

This item is a DOK 1 because it requires the student to recognize congruent shapes.

**Distractor Rationale:**

- A. The student thought as long as one side had the same measure this meant the shapes were congruent.
- B. The student confused congruent and similar figures.
- C. **Correct.** The student demonstrated an ability to identify congruent shapes after transformations.
- D. The student thought as long as one side had the same measure this meant the shapes were congruent.

**31** The list shows the number of dollars Pablo saved each week.

**4, 7, 6, 2, 10, 4**

**What is the difference between the mean and the median of these amounts?**

- A** \$0.50
- B** \$1.00
- C** \$1.50
- D** \$3.00

**Standard: 6.D.1.1** Interpret the mean, median, and mode for a set of data.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to determine the mean and median of a data set and then find the difference between the two measures of center.

**Distractor Rationale:**

- A. Correct.** The student demonstrated an ability to calculate the mean and median for a set of real-world data.
- B.** The student found the difference between the median and the mode.
- C.** The student found the difference between the mean and the mode.
- D.** The student found the difference between the range and the median.

- 32** This table shows the heights of 9 flowers in a garden.

**Flower Heights**

| <b>Flower</b> | <b>Height (inches)</b> |
|---------------|------------------------|
| A             | 2                      |
| B             | 2                      |
| C             | 2                      |
| D             | 3                      |
| E             | 3                      |
| F             | 4                      |
| G             | 6                      |
| H             | 6                      |
| I             | 18                     |

Which statement explains the measure of central tendency that will provide the **most** descriptive information about the flower heights?

- A** The mean is the best measure of central tendency for this data set because it represents the average height.
- B** The median is the best measure of central tendency for this data set because there is one value that is much greater than the other values.
- C** The mode is the best measure of central tendency for this data set because it is the most frequently occurring height.
- D** The mean, median, and mode are all equally descriptive for this data set.

**Standard: 6.D.1.2** Explain and justify which measure of center (mean, median, or mode) would provide the most descriptive information for a given set of data.

**Depth-of-Knowledge: 3**

This item is a DOK 3 because it requires the student to use reasoning to determine the best measure of central tendency for a set of data.

**Distractor Rationale:**

- A. The student thought the mean is always the most descriptive measure of central tendency.
- B. Correct. The student demonstrated an ability to identify and justify the best measure of central tendency for a data set.**
- C. The student chose because there were three 2s.
- D. The student thought all measures are equally descriptive.

**33** A bag contains 12 yellow tiles and 12 blue tiles. A student will choose one tile from the bag without looking. Which word(s) describe the probability of choosing a blue tile from the bag?

- A** likely
- B** certain
- C** impossible
- D** equally likely

**Standard: 6.D.2.1** Represent possible outcomes using a probability continuum from impossible to certain.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to understand the probability experiment described and then represent the probability of a particular outcome using words from impossible to certain.

**Distractor Rationale:**

- A. The student confused likely and equally likely.
- B. The student did not know what certain meant.
- C. The student did not know what impossible meant.
- D. Correct.** The student demonstrated an ability to represent the outcome of an event using a probability continuum from impossible to certain.

**34** These colored candies are in a bag.

**Candy in Bag**

| Color  | Number |
|--------|--------|
| blue   | 5      |
| green  | 8      |
| red    | 6      |
| yellow | 1      |

**Sarah chooses one candy from the bag without looking. Which statement is true?**

- A** It is certain that Sarah will pick a green candy.
- B** It is impossible for Sarah to pick a purple candy.
- C** Sarah is equally likely to pick a blue candy or a red candy.
- D** Sarah is less likely to pick a red candy than a yellow candy.

**Standard: 6.D.2.1** Represent possible outcomes using a probability continuum from impossible to certain.

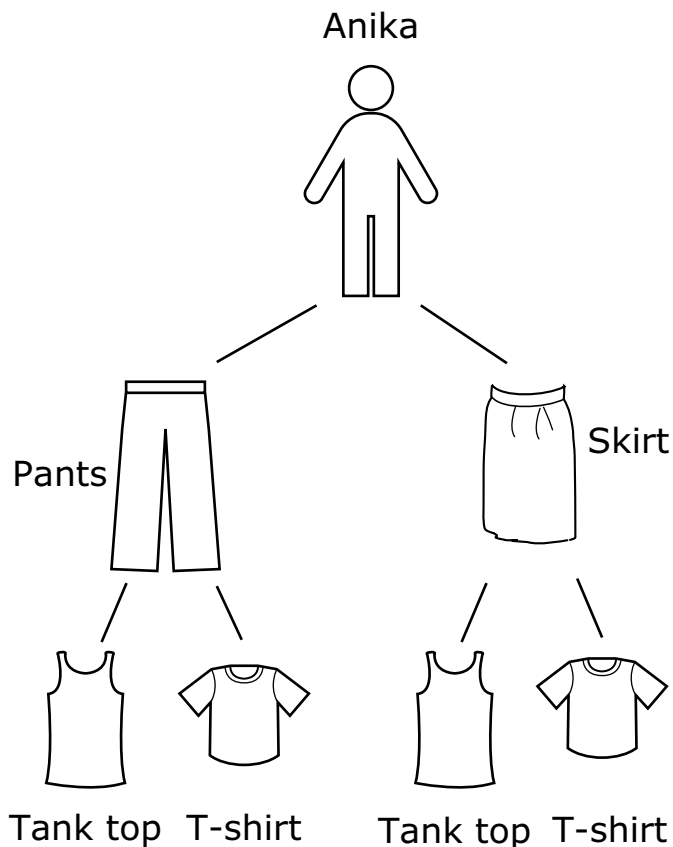
**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to understand the probability experiment described and then determine which statement is true about the possible outcomes.

**Distractor Rationale:**

- A. The student confused certain and impossible.
- B. Correct. The student demonstrated an ability to represent the possible outcomes using a probability continuum from impossible to certain.**
- C. The student saw that blue and red are very close and thought that was close enough to equal.
- D. The student reversed the relationship.

**35** Anika will choose an outfit from the clothes shown.



**Which outfit is impossible for Anika to choose?**

- A** tank top and pants
- B** tank top and skirt
- C** T-shirt and pants
- D** T-shirt and shorts

**Standard: 6.D.2.1** Represent possible outcomes using a probability continuum from impossible to certain.

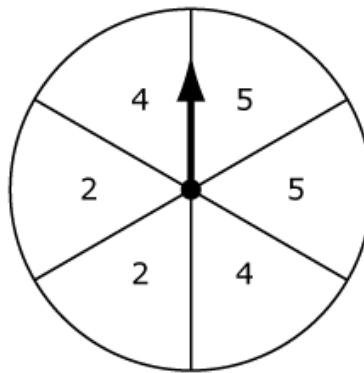
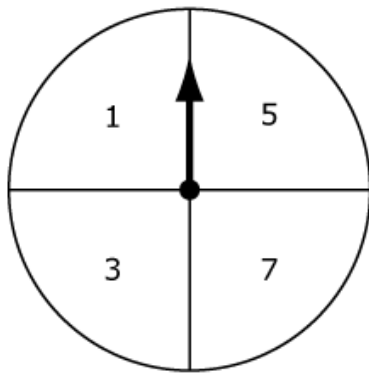
**Depth-of-Knowledge: 3**

This item is a DOK 3 because it requires the student to understand the information presented in a tree diagram and then use reasoning to determine which outcome is impossible.

**Distractor Rationale:**

- A. The student did not see that this was a possible outcome or misunderstood “impossible.”
- B. The student did not see that this was a possible outcome or misunderstood “impossible.”
- C. The student did not see that this was a possible outcome or misunderstood “impossible.”
- D. **Correct.** The student demonstrated an ability to represent an outcome from a probability situation as impossible.

Trevor spins the pointer on each of these spinners.



Select the events (left spinner, right spinner) that are members of the sample space for Trevor's spins. To select an event, click the event. To deselect the event, click it again.

(2, 2)

(1, 3)

(odd number,  
even number)

(3, 4)

(7, 1)

(odd number,  
odd number)

(5, 5)

(even number,  
even number)

**Standard: 6.D.2.2** Determine the sample space for a given experiment and determine which members of the sample space are related to certain events. Sample space may be determined by the use of tree diagrams, tables or pictorial representations.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to understand the probability experiment described and then differentiate between possible and impossible outcomes.

**Sample Distractor Rationales:**

Correct

(odd number,  
even number)

(3, 4)

(odd number,  
odd number)

(5, 5)

Incorrect

|        |                             |        |                               |
|--------|-----------------------------|--------|-------------------------------|
| (2, 2) | (1, 3)                      |        |                               |
| (7, 1) | (odd number,<br>odd number) | (5, 5) | (even number,<br>even number) |

The student chose all events that are possible with two spins on the same spinner.

|        |                             |                              |                               |
|--------|-----------------------------|------------------------------|-------------------------------|
| (2, 2) | (1, 3)                      | (odd number,<br>even number) | (3, 4)                        |
| (7, 1) | (odd number,<br>odd number) | (5, 5)                       | (even number,<br>even number) |

The student chose all events that are possible with two individual spins of either spinner.

## Cluster Items

The following sample items are part of a cluster. The cluster is presented first and then the two items that follow require use of the cluster. The two items are from different standards.

Use this information to answer the following questions.

Three expressions are shown.

|              |              |
|--------------|--------------|
| Expression 1 | $-6 + 4$     |
| Expression 2 | $1.5(9 - 7)$ |
| Expression 3 | $4.6 \div 2$ |

**37** What is the sum of Expression 2 and Expression 3?

- A 3.8
- B 5.3
- C 7.3
- D 17.3

**Standard: 6.A.2.1** Generate equivalent expressions and evaluate expressions involving positive rational numbers by applying the commutative, associative, and distributive properties and order of operations to model and solve mathematical problems.

**Depth-of-Knowledge: 2**

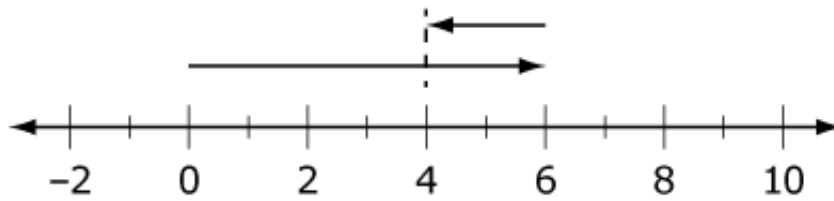
This item is a DOK 2 because it requires the student to use the order of operations and mathematical properties to find the sum of multi-step expressions.

**Distractor Rationale:**

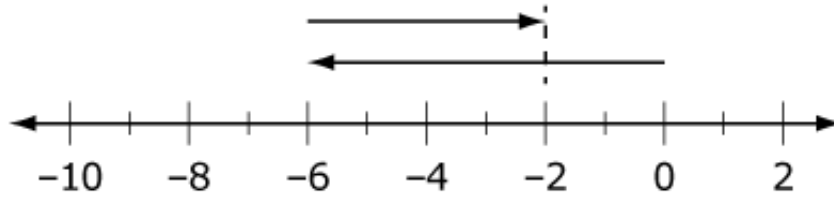
- A. The student computed  $1.5 + 2.3$ , ignoring the  $(9 - 7)$ .
- B. **Correct.** The student demonstrated an ability to evaluate an expression applying the order of operations.
- C. The student made a calculation error.
- D. The student made a calculation error.

**38 Which number line represents Expression 1?**

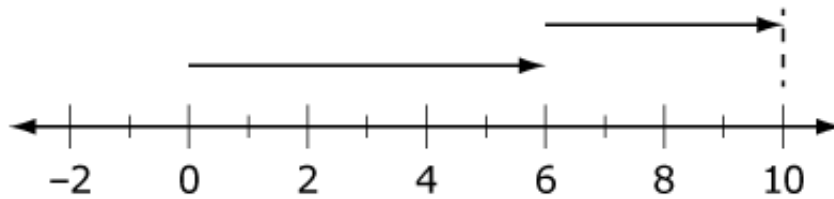
**A**



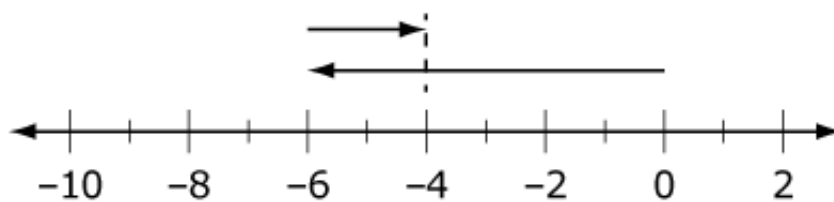
**B**



**C**



**D**



**Standard: 6.N.2.2** Illustrate addition and subtraction of integers using a variety of representations.

**Depth-of-Knowledge: 2**

This item is a DOK 2 because it requires the student to understand the addition of integers and then illustrate this addition using a number line model.

**Distractor Rationale:**

A. The student confused  $-6 + 4$  and  $6 - 2$ .

B. **Correct.** The student demonstrated an ability to illustrate an addition expression on the number line.

C. The student confused  $-6 + 4$  and  $6 + 4$ .

D. The student confused  $-6 + 4$  and  $-6 + 2$ .